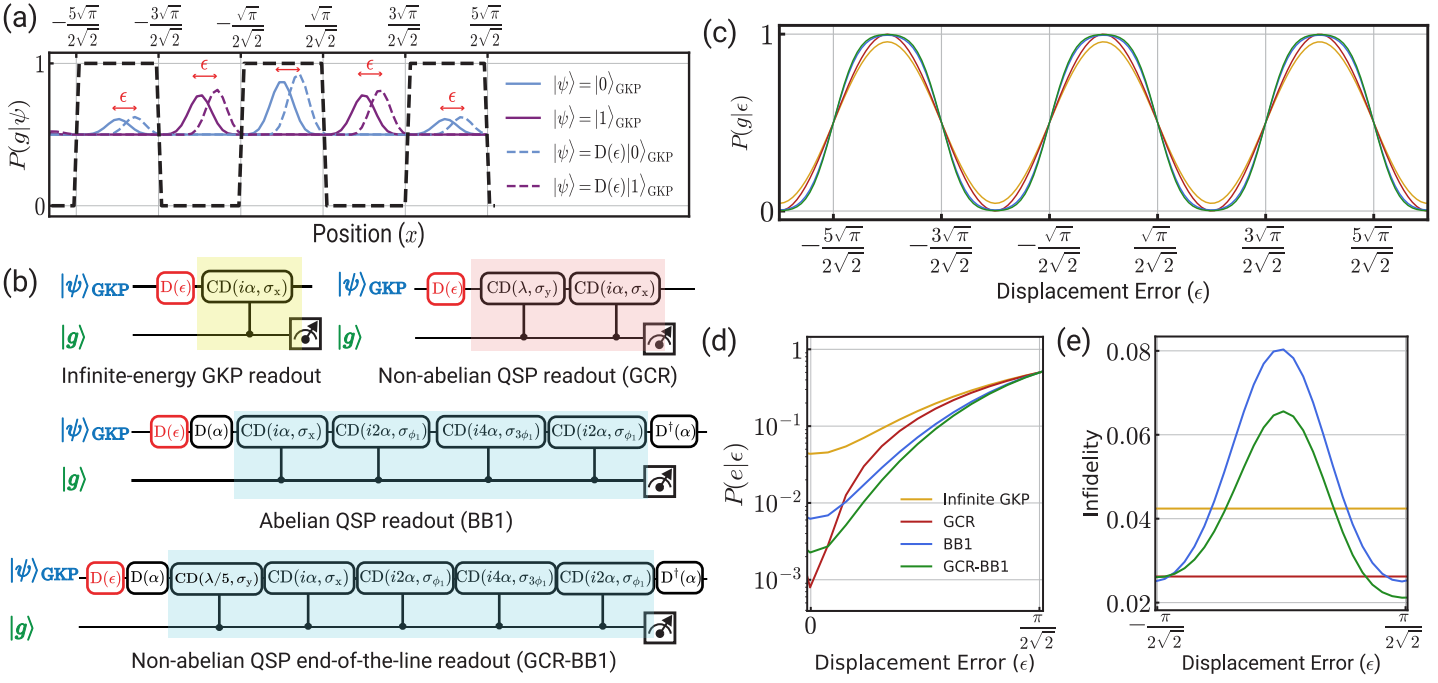
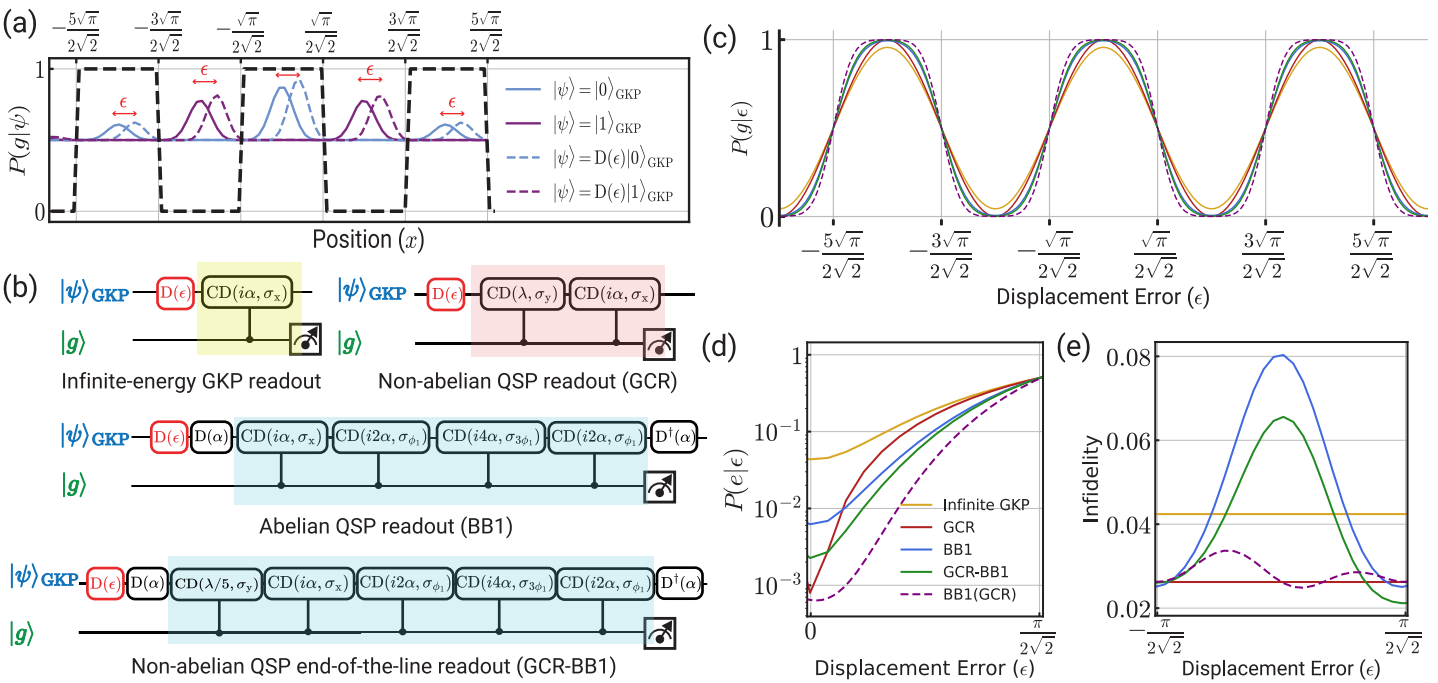

(a)
$-\frac{5\sqrt{\pi}}{2\sqrt{2}}$ $-\frac{3\sqrt{\pi}}{2\sqrt{2}}$ $-\frac{\sqrt{\pi}}{2\sqrt{2}}$ $\frac{\sqrt{\pi}}{2\sqrt{2}}$ $\frac{3\sqrt{\pi}}{2\sqrt{2}}$ $\frac{5\sqrt{\pi}}{2\sqrt{2}}$
$P(g|\psi)$
1
0
ϵ
$|\psi\rangle = |0\rangle_{\mathrm{GKP}}$
$|\psi\rangle = |1\rangle_{\mathrm{GKP}}$
$|\psi\rangle = \mathrm{D}(\epsilon)|0\rangle_{\mathrm{GKP}}$
$|\psi\rangle = \mathrm{D}(\epsilon)|1\rangle_{\mathrm{GKP}}$
Position (x)
(b)
$|\psi\rangle$ GKP
$|g\rangle$
D(ϵ)
CD($i\alpha, \sigma_x$)
CD(λ, σ_y)
Infinite-energy GKP readout
Non-abelian QSP readout (GCR)
D(α)
CD($i2\alpha, \sigma_{\phi_1}$)
CD($i4\alpha, \sigma_{3\phi_1}$)
D†(α)
Abelian QSP readout (BB1)
CD($\lambda/5, \sigma_y$)
Non-abelian QSP end-of-the-line readout (GCR-BB1)
(c)
$P(g|\epsilon)$
1
0
$-\frac{5\sqrt{\pi}}{2\sqrt{2}}$ $-\frac{3\sqrt{\pi}}{2\sqrt{2}}$ $-\frac{\sqrt{\pi}}{2\sqrt{2}}$ $\frac{\sqrt{\pi}}{2\sqrt{2}}$ $\frac{3\sqrt{\pi}}{2\sqrt{2}}$ $\frac{5\sqrt{\pi}}{2\sqrt{2}}$
Displacement Error (ϵ)
(d)
$P(e|\epsilon)$
1
10^{-1}
10^{-2}
10^{-3}
Infinite GKP
GCR
BB1
GCR-BB1
0
$\frac{\pi}{2\sqrt{2}}$
Displacement Error (ϵ)
(e)
Infidelity
0.08
0.06
0.04
0.02
$-\frac{\pi}{2\sqrt{2}}$
$\frac{\pi}{2\sqrt{2}}$
Displacement Error (ϵ)


(a)
$P(g|\psi)$
Position (x)
$|\psi\rangle = |0\rangle_{\mathrm{GKP}}$
$|\psi\rangle = |1\rangle_{\mathrm{GKP}}$
$|\psi\rangle = \mathrm{D}(\epsilon)|0\rangle_{\mathrm{GKP}}$
$|\psi\rangle = \mathrm{D}(\epsilon)|1\rangle_{\mathrm{GKP}}$
(b)
Infinite-energy GKP readout
Non-abelian QSP readout (GCR)
Abelian QSP readout (BB1)
Non-abelian QSP end-of-the-line readout (GCR-BB1)
(c)
$P(g|\epsilon)$
Displacement Error (ϵ)
(d)
$P(e|\epsilon)$
Infinite GKP
GCR
BB1
GCR-BB1
BB1(GCR)
Displacement Error (ϵ)
(e)
Infidelity
Displacement Error (ϵ)